SIMATS ENGINEERING

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Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 - ASSESSMENT TOOL 2

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO1: Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.

(BL-1, BL-2, BL-3)

Assessment Tool Weightage:

QUESTIONS: 10 **Total Marks: 100**

Annexure A

Concept Mapping

Code of Conduct:

I P.Sai Ganesh Reddy(192525111) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Sai Ganesh Reddy

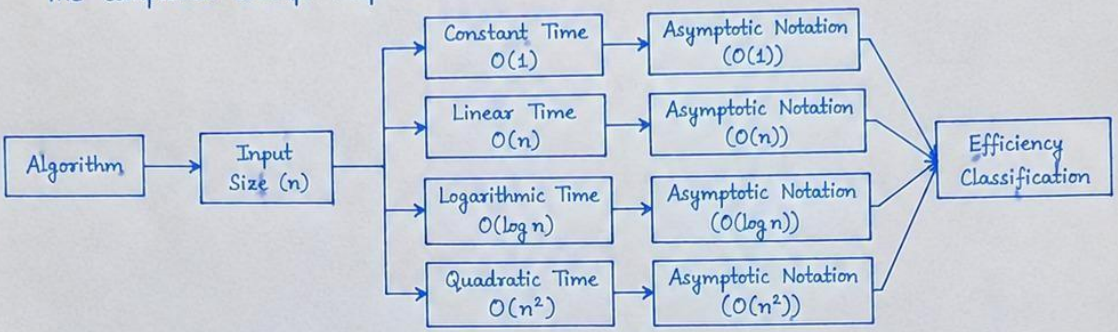
RUBRICS FOR CALCULATION

Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

9. Consider the concept map: Algorithm → Input Size → Constant Time, Linear Time, Logarithmic Time, Quadratic Time → Efficiency Classification. Explain how changes in input size lead to different growth patterns and how these patterns are classified. Extend the map by linking Asymptotic Notation to each complexity class. Justify why logarithmic and linear algorithms are generally more scalable than quadratic algorithms. Interpret how this concept map helps in classifying algorithms based on their order of growth.

Answer:

The completed concept map is shown below:



1. **How input size leads to different growth patterns:**
   - As the input size (n) increases, the number of operations required by an algorithm changes according to its growth pattern.
   - Different algorithms have different rates of increase in running time, which are classified into constant, linear, logarithmic and quadratic growth patterns.

2. **Classification using Asymptotic Notation:**
   - Each complexity class is expressed using asymptotic notation (Big-O) to show the order of growth of the running time.
   - The map links each time complexity class to its corresponding Big-O notation ($O(1)$, $O(n)$, $O(\log n)$, $O(n^2)$).

3. **Why Logarithmic and Linear are more scalable than quadratic:**
   - Logarithmic ($O(\log n)$) and linear ($O(n)$) algorithms grow much slower than quadratic ($O(n^2)$) as n increases.
   - This means they remain efficient even for large inputs, whereas quadratic algorithms become significantly slower with increasing input size.

4. **Interpretation:**
   - The concept map provides a clear and organized structure to classify algorithms based on their order of growth.
   - By linking input size → growth pattern → asymptotic notation → classification, it helps in comparing and selecting the most efficient algorithm for a given problem.

Conclusion:

Thus, the concept map helps in classifying algorithms systematically by showing how input size affects the growth pattern and how asymptotic notation expresses it, making it easier to identify the most efficient algorithm for large inputs.